

1 Virtual displacements and work

- Forces that are **opposite** in direction to its displacement result in **negative work**.

Internal forces come in action-reaction pairs and hence are equal in magnitude but opposite in direction.

1.1 Virtual displacements

$\delta W = 0 \Rightarrow P_1 \delta x_1 + P_2 \delta x_2 + \dots + P_n \delta x_n = 0$

1.2 Virtual work

$\delta W = \delta U$
 $P_1 \delta x_1 + P_2 \delta x_2 + \dots + P_n \delta x_n = F_1 \delta e_1 + F_2 \delta e_2 + \dots + F_m \delta e_m$

1.3 Steps

- Set up a coordinate system with the origin at a **fixed point**
- Assign displacement coordinates to each force
- Write out $\delta W = 0$ (virtual displacements) or $\delta W = \delta U$ (virtual work)
- Express all coordinate systems to a common variable (usually given) using $\delta y = \frac{dy}{dx} \delta x$
- Drop the common infinitesimal variable (δx)
- Solve for the applied force

2 Virtual complementary work

$\delta W^* = \delta U^*$
 $\Delta_1 \delta F_1 + \Delta_2 \delta F_2 + \dots + \Delta_n \delta F_n = e_1 \delta F_1 + e_2 \delta F_2 + \dots + e_m \delta F_m$

2.1 Steps

- Write $\delta W^* = \delta U^*$
- Drop unwanted deflection terms (Δ_n), and leave only the desired deflection term from the question
- Replace e_m with $\frac{F_m}{k_m}$
- Find the spring forces $F_1, F_2, \dots, F_m$ using moments or force balance
- Find the virtual forces $\delta F_1, \delta F_2, \dots, \delta F_m$ using virtual force or moment balance
 - Apply a virtual force δP at the desired deflection point **only**
 - There are no external virtual forces anywhere else
- Find $\delta F_1, \delta F_2, \dots, \delta F_m$ in terms of δP
- Substitute the results from steps 4 and 5 into the step 3 equation, and solve for the desired deflection

3 Unit load method

$\Delta = \delta U^*$

3.1 Equilibrium equations

$\sum F_x = 0, \sum F_y = 0, \sum M = 0$

3.2 Degree of indeterminacy

$n = N_{\text{unknowns}} - 3$

3.3 Steps

- Write:
 - How to decide where does a segment end?
 - Force changes
 - Beam changes (E, I or cross-section)
- Solve for unknown reaction forces
- Find internal moments of each segment $M_1, M_2, \dots, M_n$
 - Take a cut at the segment
 - Wherever the cut is made, draw the internal moment, like M_1
 - Take $\sum M = 0$ about the cutting point to find M_1
- Perform virtual load (for linear displacement) or moment (for angular displacement) analysis
 - Remove all forces
 - For statically indeterminate structures, replace the unknown force or moment with a unit load or moment.
 - Apply unit load (1 N) or moment (1 Nm) at the point of interest (deflection point)
 - Find unknown reaction forces
 - Find internal virtual moment in each segment $m_1, m_2, \dots, m_n$
 - The internal virtual moments must be in the **same direction** as the actual internal moments.
- Substitute $M_1, M_2, \dots, M_n$, and $m_1, m_2, \dots, m_n$ into Δ and evaluate the integral

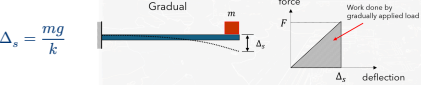
For statically indeterminate structures, find a point that has **zero** deflection in one direction, like a slot, then apply the unit load method with the unit load in the same direction to find the unknown internal force. Then proceed with solving the question using the found force.

3.4 Useful equations

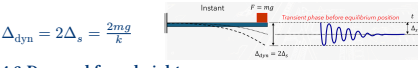
$s = r\theta \Rightarrow ds = r d\theta$

4 Impact loads

4.1 Gradual release



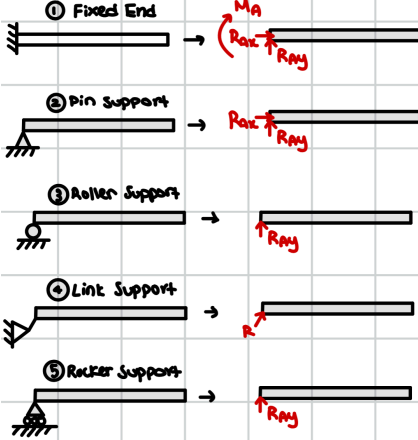
4.2 Instant release



4.3 Dropped from height



4.4 Types of supports and their reaction forces



5 Brittle fracture

5.1 Strain energy release rate

- $G = \frac{\pi \sigma^2 a}{2E}$
- a is **half** the crack length
 - $k = 1$ for a thin plate (plane stress)
 - $k = (1 - \nu^2)$ for a thick block (plane strain)

5.2 Stress intensity factor

$K_I = Y \sigma \sqrt{\pi a}$
 $\sigma_{tip} = \frac{K_I}{\sqrt{2\pi a}}$
 σ is the geometric correction factor, which is usually 1.

5.2.1 Geometric correction factor

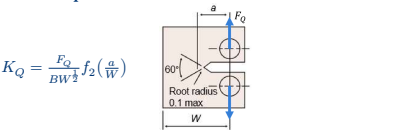
Finite width plate: $Y = \left(\frac{W}{\pi a} \tan \frac{\pi a}{W} \right)^{\frac{1}{2}}$
Double-edge crack: $Y = \left(\frac{W}{\pi a} \tan \frac{\pi a}{W} + \frac{0.2W}{\pi a} \sin \frac{\pi a}{W} \right)^{\frac{1}{2}}$
Internal penny crack: $Y = \frac{2}{\pi}$

5.3 Variation of fracture toughness

5.3.1 Bending



5.3.2 Compact tension



5.3.3 Valid test criterion

$\frac{F_{max}}{Q} < 1.1$

5.3.4 Valid KIC criteria

$B, a, (W - a) \geq 2.5 \left(\frac{K_{Ic}}{\sigma_y} \right)^2$

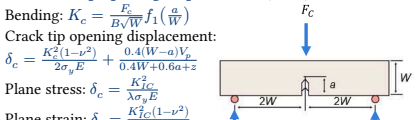
$0.45 < \frac{a}{W} < 0.55$

6 Crack tip plasticity

6.1 Plastic zone correction

$K = \sigma \sqrt{\pi(a + r_p)}$
Plane stress $r_p = \frac{1}{2\pi} \left(\frac{K}{\sigma_y} \right)^2$ **Plane strain** $r_p = \frac{1}{6\pi} \left(\frac{K}{\sigma_y} \right)^2$

6.2 Crack tip opening displacement (COD) method



6.3 J-integral

$J = \int_C \{ w dy - t \left(\frac{\partial u}{\partial x} \right) ds, w = \int_0^\epsilon \sigma d\epsilon$
 w is the strain energy density, u is the displacement vector
 T is the traction vector, $T = s \cdot n$
For linear elastic materials: $J_c = G_c, \therefore K_c = \sqrt{E J_c}$

$J = -\frac{1}{B} \frac{\partial U}{\partial a} = \frac{2U}{B(W-a)}$
 U is the strain energy, a is **half** the crack length
 B is the specimen thickness, W is the specimen width

7 Cyclic stresses

Stress amplitude: $S_a = \sigma_a = \frac{1}{2}(\sigma_{max} - \sigma_{min})$
Mean stress: $S_m = \sigma_m = \frac{1}{2}(\sigma_{max} + \sigma_{min})$
Stress ratio: $R = \frac{\sigma_{min}}{\sigma_{max}}$
Stress range: $S_r = \sigma_r = \sigma_{max} - \sigma_{min}$

For the completely reversed stress cycle, only the tensile part contributes to crack growth, so the stress range for **crack growth** is only the tensile part of the cycle, i.e.

$S_r = \sigma_r = \sigma_{max} - 0 = \sigma_{max}$
 S_e is the endurance limit, S_u is the ultimate tensile strength.
 $S_e = 0.5 S_u$ for steels with $S_u < 690$ MPa
 $S_e = 0.4 S_u$ for aluminium alloys with $S_u < 131$ MPa
 $S_e = 0.4 S_u$ for copper alloys with $S_u < 96.5$ MPa

Safety factor against fatigue failure = $\frac{\text{Endurance limit, } S_e}{\text{Stress amplitude, } S_a}$
SF = $\frac{S_e}{S_a}, \sigma = \frac{My}{I}, I = \frac{\pi d^4}{64}, y = \frac{D}{2}$

$S_e = S'_e C_{size} C_{load} C_{surface-finish}$
 S_e is the modified endurance limit
 S'_e is the test specimen endurance limit.
For circular rods:

$C_{size} = \begin{cases} 1.0 & \text{if } d \leq 8 \text{ mm for bending} \\ 1.189d^{-0.097} & \text{if } 8 \text{ mm} < d \leq 250 \text{ mm for torsional load} \\ 1.0 & \text{for axial load} \end{cases}$
 $C_{load} = \begin{cases} 1.0 & \text{for bending loads in the component} \\ 0.7 & \text{for axial loads in the component} \\ 0.577 & \text{for torsional loads in the component} \end{cases}$

$S_e = \frac{S'_e C_{size} C_{load} C_{surface-finish}}{K_f}$
 K_f is the fatigue strength reduction factor
 $q = \frac{K_f - 1}{K_f} \Rightarrow K_f = q(K_t - 1) + 1$

- q is the notch sensitivity factor, $0 \leq q \leq 1$
- K_t is the theoretical elastic stress concentration factor

7.1 Stress criteria

The below criteria are for **NO** fatigue failure.

Goodman $\frac{S_a}{S_e} + \frac{S_m}{S_u} \leq 1$
Soderberg $\frac{S_a}{S_e} + \left(\frac{S_m}{S_u} \right)^2 \leq 1$ $\frac{S_a \times SF}{S_e} + \left(\frac{S_m}{S_u} \right)^2 \leq 1$
 $S_r = \sigma_{max} - \sigma_{min}$
 $\frac{S_a \times SF}{S_e} + \frac{S_m}{S_u} \leq 1$

If mean stress is 0: SF = $\frac{S_e}{S_a} \Rightarrow \frac{S_a \times SF}{S_e} = 1$
If the mean stress is non-zero, look at the right side of the equation.

7.2 Miner's rule for cumulative fatigue damage

$D = \frac{n_1}{N_1} + \frac{n_2}{N_2} + \dots + \frac{n_k}{N_k} = 1, N = \frac{1}{D}$
 D is the damage, so if $D \geq 1$, the material will fail
 n_k number of cycles of operation at stress level S_k
 N_k number of cycles to failure at stress level S_k

7.3 Crack growth

Fatigue crack growth rate: $\frac{da}{dN}$
Stress intensity factor range (ΔK):
 $\Delta K = K_{max} - K_{min} = Y S_r \sqrt{\pi a} = Y^m S_r^m \pi^{\frac{m}{2}} a^{\frac{m}{2}}$
Paris law: $\frac{da}{dN} = C (\Delta K)^m = C (Y^m S_r^m \pi^{\frac{m}{2}} a^{\frac{m}{2}})^m$

N_f is the number of fatigue cycles, **all stresses in MPa** for N_f
 m is a coefficient, and a is half the crack length
Stress ratio: $R = \frac{K_{min}}{K_{max}} = \frac{S_{min}}{S_{max}}$

Forman equation: $\frac{da}{dN} = \frac{C (\Delta K)^m}{(1-R) K_{th} - \Delta K}$
Walker equation: $\frac{da}{dN} = C \left(\frac{\Delta K}{(1-R)^{\frac{1}{1-\gamma}}} \right)^m$

8 Impact testing

Impact strength (energy absorbed): $E = mgH_{\text{before}} - mgH_{\text{after}}$

9 Stiffness formulas

9.1 General

$k = \frac{F}{\delta}$

F is the force

δ is the deflection

9.2 Torsional stiffness

$k = \frac{EI}{l}$

- E is the modulus of elasticity, and l is the total length
- I is the moment of inertia of the cross-sectional area

9.3 Axial force on a bar

$k = \frac{EA}{l}$

A is the cross-sectional area

9.4 Torque on a bar

$k = \frac{GJ}{l}$

- G is the shear modulus or modulus of rigidity
- J is the moment of inertia

9.5 Force on the end of cantilever beam

$k = \frac{3EI}{l^3}$

Note that k is at the position of the load.

9.6 Force in the middle of a bar

$k = \frac{48EI}{l^3}$

Note that k is at the position of the load.

9.7 Springs in series

$k_1, k_2 \Rightarrow k_{eff} = \frac{1}{\frac{1}{k_1} + \frac{1}{k_2}}$

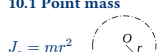
9.8 Springs in parallel

$k_1, k_2 \Rightarrow k_{eff} = k_1 + k_2$

10 Moment of inertia formulas

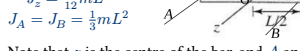
10.1 Point mass

$J_o = mr^2$



10.2 Bar

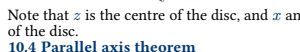
$J_z = \frac{1}{12} mL^2$
 $J_A = J_B = \frac{1}{3} mL^2$



Note that z is the centre of the bar, and A and B are at the ends of the bar.

10.3 Disc

$J_z = \frac{1}{2} mR^2$
 $J_x = J_y = \frac{1}{4} mR^2$



Note that z is the centre of the disc, and x and y are at the edge of the disc.

10.4 Parallel axis theorem

$J = J_G + Md^2$
 J is the moment of inertia needed
 J_G is the moment of inertia about its own centre of gravity
 M is the mass of the body
 d is the distance between the axes

11 Vibrations

Angular motion: $\theta = \Theta \sin(\omega t - \phi)$

Rectilinear motion: $x = X \sin(\omega t - \phi)$

Frequency: $f = \frac{1}{T}$ (Hz)

Angular frequency: $\omega = 2\pi f$ (rad s⁻¹)

Equation of motion: $J_o \ddot{\theta} = \sum M_o$

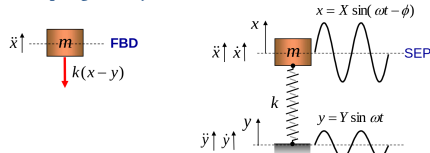
Natural frequency:

$\omega_n = \sqrt{\frac{k}{m}}$ (rad s⁻¹), $f_n = \frac{\omega_n}{2\pi}$ (Hz)

Natural period: $\tau_n = \frac{1}{f_n}$

$\theta = \Theta \sin(\omega_n t + \phi)$

11.2 Spring mass system



Equation of motion: $m\ddot{\theta} = \sum F$

Natural frequency:

$\omega_n = \sqrt{\frac{k}{m}}$ (rad s⁻¹), $f_n = \frac{\omega_n}{2\pi}$ (Hz)

Natural period: $\tau_n = \frac{1}{f_n}$

$x = X \sin(\omega_n t + \phi)$

11.2.1 Harmonic force excitation

$F(t) = F_0 \sin \omega t$

Newton's second law: $m\ddot{x} = \sum F$

Equation of motion: $m\ddot{x} + kx = F_0 \sin \omega t$

Steady state response:

$x = X \sin(\omega t - \phi)$ $\ddot{x} = -\omega^2 X \sin(\omega t - \phi)$

Phase: $\phi = 0$

Amplitude: $X = \frac{F_0}{k - m\omega^2} = \frac{F_0}{k \left(1 - \left(\frac{\omega}{\omega_n} \right)^2 \right)}$

11.2.2 Rotating unbalance

Newton's second law: $M\ddot{x} = \sum F$

M is the total mass inclusive of m .

Equation of motion: $M\ddot{x} + kx = me\omega^2 \sin \omega t$

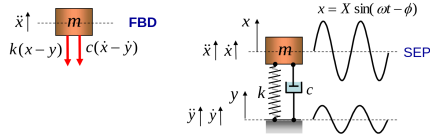
Steady state solution: $x_p = X \sin(\omega t - \phi)$

Amplitude: $X = \frac{me\omega^2}{k - M\omega^2}$

Magnification factor: $\frac{MX}{me} = \frac{\left(\frac{\omega}{\omega_n} \right)^2}{1 - \left(\frac{\omega}{\omega_n} \right)^2}$

Phase: $\phi = 0$

11.3 Spring mass damper system



Equation of motion: $m\ddot{x} + c\dot{x} + kx = 0$

Critical damping constant: $c_c = 2\sqrt{2km} = 2m\omega_n = \frac{2k}{\omega_n}$

11.3.1 Harmonic force excitation

$F(t) = F_0 \sin \omega t$

Newton's second law: $m\ddot{x} = \sum F$

Equation of motion: $m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$

Steady state response:

$x = X \sin(\omega t - \phi)$ $\ddot{x} = -\omega^2 X \sin(\omega t - \phi)$

Amplitude: $X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}} = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2} \right)^2 + \left(\frac{2c\omega}{k} \right)^2}}$

Phase: $\phi = \arctan \frac{c\omega}{k - m\omega^2} = \arctan \frac{\frac{2c\omega}{k}}{1 - \frac{\omega^2}{\omega_n^2}}$

At resonance, $\omega = \omega_n$, amplitude: $X_{res} = \frac{F_0}{2k\zeta}$

11.3.2 Rotating unbalance

Newton's second law: $M\ddot{x} = \sum F$

M is the total mass inclusive of m .

Equation of motion: $M\ddot{x} + kx = me\omega^2 \sin \omega t$

Steady state solution: $x_p = X \sin(\omega t - \phi)$

Amplitude: $X = \frac{me\omega^2}{\sqrt{(k - M\omega^2)^2 + (c\omega)^2}}$

$\phi = \arctan \frac{c\omega}{k - M\omega^2} = \arctan \frac{\frac{2c\omega}{k}}{1 - \frac{\omega^2}{\omega_n^2}}$

Magnification factor: $\frac{MX}{me} = \frac{\left(\frac{\omega}{\omega_n} \right)^2}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2} \right)^2 + \left(\frac{2c\omega}{k} \right)^2}}$

11.4 Shaft-disc system

Equation of motion: $J_o \ddot{\theta} = \sum M_o$

Natural frequency:

$\omega_n = \sqrt{\frac{k}{m}}$ (rad s⁻¹), $f_n = \frac{\omega_n}{2\pi}$ (Hz)

Torsional stiffness of the shaft:

$k_o = \frac{GJ}{L}$ (Nm rad⁻¹)

L is the length of the shaft

J is the polar moment of inertia, $J = \frac{\pi d^4}{32}$

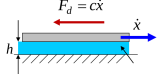
Natural period: $\tau_n = \frac{1}{f_n}$

11.5 Viscous damping

- $F_d = c\dot{x}$ $c = \frac{\mu A}{h}$
- F_d is the viscous force on the mass (N)
 - μ is the dynamic viscosity (Pa s)
 - A is the contact area (m²)
 - h is the film thickness (m)

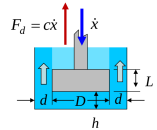
Parallel-plate (shear) damper

$$c = \frac{\mu A}{h}$$
$$F_d = c\dot{x}$$



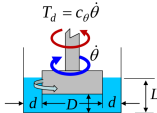
Axial damper (dashpot)

$$c = \frac{3\pi\mu D^3 L}{4d^3} \left(1 + \frac{2d}{D}\right)$$
$$F_d = c\dot{x}$$



Torsion damper

$$c_\theta = \frac{\pi\mu D^3(L-h)}{4d} + \frac{\pi\mu D^4}{32h}$$
$$T_d = c_\theta\dot{\theta}$$



11.6 Equations of motion

- $m\ddot{x} + c\dot{x} + kx = 0$
- $J_o\ddot{\theta} + c_\theta\dot{\theta} + k_\theta\theta = 0$
- m and J_o are the effective masses
 - c and c_θ are the damping constants
 - k and k_θ are the effective stiffness

11.7 Natural frequency

$$\omega_n = \sqrt{\frac{\text{Effective stiffness}}{\text{Effective mass}}} = \sqrt{\frac{k}{m}} = \sqrt{\frac{k_\theta}{J_o}}$$
$$f_n = \frac{1}{2\pi} \sqrt{\frac{\text{Effective stiffness}}{\text{Effective mass}}} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{k_\theta}{J_o}}$$

11.8 Under-damped frequency

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

11.9 Critical damping constant

$$c_c = 2\sqrt{km} = 2m\omega_n = \frac{2k}{\omega_n}$$
$$(c_\theta)_c = 2\sqrt{k_\theta J_o} = 2J_o\omega_n = \frac{2k_\theta}{\omega_n}$$

11.10 Damping ratio

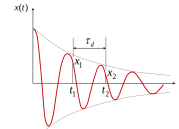
$$\zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{km}} = \frac{c}{2m\omega_n} = \frac{c\omega_n}{2k}$$
$$\zeta = \frac{c}{(c_\theta)_c} = \frac{c}{2\sqrt{k_\theta J_o}} = \frac{c}{2J_o\omega_n} = \frac{c\omega_n}{2k_\theta}$$

$$\zeta = \sqrt{\frac{\delta^2}{4\pi^2 + \delta^2}}$$

11.11 Logarithmic decrement

$$\delta = \ln \frac{x_1}{x_2} = \frac{2\pi\zeta}{\sqrt{1 - \zeta^2}}$$

- x_1 is the amplitude of the first cycle
- x_2 is the amplitude of the second cycle



11.12 Static analysis

$\sum F = 0$
 $\sum M = 0$

Static analysis is required when there is a spring with an unknown extension in static equilibrium, like it is counteracting the weight of the object.

11.13 Dynamic analysis

Newton's 2nd law:

$$m\ddot{\theta} = \sum F \quad J_o\ddot{\theta} = \sum M_o$$

11.14 Undamped free vibrations

11.14.1 Equation of motion

$$m\ddot{x} + kx = 0$$

11.14.2 Displacement, velocity and acceleration

$$x = X \sin(\omega_n t + \phi)$$
$$\dot{x} = \omega_n X \cos(\omega_n t + \phi)$$
$$\ddot{x} = -\omega_n^2 X \sin(\omega_n t + \phi)$$

11.14.3 Amplitude

$$X = \sqrt{x_0^2 + \left(\frac{\dot{x}_0}{\omega_n}\right)^2}$$

11.14.4 Phase

$$\phi = \arctan\left(\frac{\dot{x}_0\omega_n}{x_0}\right) + 180^\circ$$

11.15 Undamped forced vibrations

11.15.1 Equation of motion

$$m\ddot{x} + kx = F_0 \sin \omega t$$

11.15.2 Displacement

$$x = x_h + x_p$$
$$x = A \cos \omega_n t + B \sin \omega_n t + X \sin(\omega t - \phi)$$
$$x = X_p \sin(\omega_n t + \phi_h) + X_p \sin(\omega t - \phi_p)$$

11.15.3 Steady-state solution

$$x_p = X \sin(\omega t - \phi)$$

11.15.4 Amplitude

$$X = \frac{\frac{F_0}{k}}{1 - \left(\frac{\omega}{\omega_n}\right)^2} = \frac{F_0}{k - m\omega^2}$$
$$\frac{X}{\left(\frac{F_0}{k}\right)} = \frac{1}{1 - \left(\frac{\omega}{\omega_n}\right)^2}$$

11.15.5 Phase

$$\phi = 0$$

11.16 Damped free vibrations

11.16.1 Equation of motion

$$m\ddot{x} + c\dot{x} + kx = 0$$

11.16.2 Under-damped

Damping ratio: $\zeta < 1$

$$x = e^{-\zeta\omega_n t} X \sin(\omega_d t + \phi)$$
$$\dot{x} = -\zeta\omega_n e^{-\zeta\omega_n t} X \sin(\omega_d t + \phi) + e^{-\zeta\omega_n t} \omega_d X \cos(\omega_d t + \phi)$$
$$= -\zeta\omega_n x + e^{-\zeta\omega_n t} \omega_d X \cos(\omega_d t + \phi)$$
$$\ddot{x} = e^{-\zeta\omega_n t} X \sin(\omega_d t + \phi) (\zeta^2 \omega_n^2 - \omega_d^2)$$
$$- 2e^{-\zeta\omega_n t} \zeta \omega_d \omega_n X \cos(\omega_d t + \phi)$$
$$= x (\zeta^2 \omega_n^2 - \omega_d^2) - 2e^{-\zeta\omega_n t} \zeta \omega_d \omega_n X \cos(\omega_d t + \phi)$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

11.16.3 Critically damped

Damping ratio: $\zeta = 1$

$$x = (C_1 + C_2 t) e^{-\omega_n t}$$
$$\dot{x} = C_2 e^{-\omega_n t} - \omega_n (C_1 + C_2 t) e^{-\omega_n t}$$
$$= C_2 e^{-\omega_n t} - \omega_n x$$
$$\ddot{x} = \omega_n^2 (C_1 + C_2 t) e^{-\omega_n t} - 2C_2 \omega_n e^{-\omega_n t}$$
$$= x - 2C_2 \omega_n e^{-\omega_n t}$$

11.16.4 Over-damped

Damping ratio: $\zeta > 1$

$$x = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$
$$\dot{x} = C_1 r_1 e^{r_1 t} + C_2 r_2 e^{r_2 t}$$
$$\ddot{x} = C_1 r_1^2 e^{r_1 t} + C_2 r_2^2 e^{r_2 t}$$
$$r_1 = -\zeta\omega_n + \omega_n \sqrt{\zeta^2 - 1}$$
$$r_2 = -\zeta\omega_n - \omega_n \sqrt{\zeta^2 - 1}$$

11.16.5 Phase

$$\phi = \arctan\left(\frac{c\omega}{k - m\omega^2}\right) = \arctan\left(\frac{\frac{2\zeta\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}\right)$$

11.17 Damped forced vibrations

11.17.1 Equation of motion

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$$

- Unbalanced mass: $M\ddot{x} + c\dot{x} + kx = m\omega^2 \sin \omega t$
- M is the effective mass, adding all the masses together
 - m is the unbalanced mass
 - e is the eccentricity

11.17.2 Displacement

$$x = e^{-\zeta\omega_n t} (A \cos \omega_d t + B \sin \omega_d t) + X \sin(\omega t - \phi)$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

11.17.3 Steady-state solution

$$x_p = X \sin(\omega t - \phi)$$

11.17.4 Amplitude

$$X = \frac{\frac{F_0}{k}}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

At resonant frequency: $X_{\text{res}} = \frac{F_0}{\sqrt{(k - m\omega_n^2)^2 + (c\omega_n)^2}}$

$$\text{Unbalanced mass: } X = \frac{\frac{F_0}{k}}{\sqrt{(k - M\omega^2)^2 + (c\omega)^2}}$$

11.17.5 Magnification factor or ratio

$$\frac{X}{\left(\frac{F_0}{k}\right)} = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(\frac{2\zeta\omega}{\omega_n}\right)^2}}$$
$$\frac{X}{\left(\frac{F_0}{k}\right)} = \frac{1}{2\zeta} \text{ when } \omega = \omega_n$$

$$\text{Unbalanced mass: } \frac{MX}{me} = \frac{\left(\frac{\omega_n}{\omega}\right)^2}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(\frac{2\zeta\omega}{\omega_n}\right)^2}}$$

$$\frac{MX}{me} = \frac{1}{2\zeta} \text{ when } \omega = \omega_n$$

11.18 Force transmission ratio

Without damping:

$$\text{TR} = \left| \frac{F_T}{F_0} \right| = \frac{kX}{F_0} = \frac{k}{|k - m\omega^2|} = \frac{1}{\left| 1 - \left(\frac{\omega}{\omega_n}\right)^2 \right|}$$

With damping:

$$\text{TR} = \frac{F_T}{F_0} = \frac{X\sqrt{k^2 + (c\omega)^2}}{F_0} = \frac{\sqrt{k^2 + (c\omega)^2}}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$
$$= \frac{\sqrt{1 + \left(\frac{2\zeta\omega}{\omega_n}\right)^2}}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(\frac{2\zeta\omega}{\omega_n}\right)^2}}$$

For torsional vibrations, replace F_T and F_0 in the above equations with M_T and M_0 respectively.

11.19 Transient vibrations

11.19.1 Due to initial conditions

Without damping:

$$m\ddot{x} + kx = F_0 \sin \omega t$$
$$x = x_h(t) + x_p(t) = A \cos \omega_n t + B \sin \omega_n t + X \sin(\omega t - \phi)$$

With damping:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$$
$$x = x_h(t) + x_p(t)$$
$$= e^{-\zeta\omega_n t} (A \cos \omega_d t + B \sin \omega_d t) + X \sin(\omega t - \phi)$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

11.19.2 Spring mass system under step loading

$$x = \frac{F_0}{k} (1 - \cos \omega_n t)$$

11.19.3 Spring mass system with sudden load

$$x = \frac{mg}{k} (1 - \cos \omega_n t)$$

11.19.4 Drop test without damping

$$x(t) = \sqrt{\left(\frac{mg}{k}\right)^2 + \left(\frac{v_0}{\omega_n}\right)^2} \sin(\omega_n t + \phi) + \frac{mg}{k}$$
$$\phi = \arctan\left(\frac{-mg\omega_n}{kv_0}\right)$$
$$\dot{x}(t) = \omega_n \sqrt{\left(\frac{mg}{k}\right)^2 + \left(\frac{v_0}{\omega_n}\right)^2} \cos(\omega_n t + \phi)$$

$$\ddot{x}(t) = -\omega_n^2 \sqrt{\left(\frac{mg}{k}\right)^2 + \left(\frac{v_0}{\omega_n}\right)^2} \sin(\omega_n t + \phi)$$

$$\Delta_{\text{max}} = \frac{g}{\omega_n^2} = \frac{mg}{k}$$

$$\frac{-\ddot{x}_{\text{max}}}{g} = \sqrt{1 + \frac{2h}{\Delta_{\text{max}}}}$$

11.19.5 Drop test with damping

$\zeta > 1$ $x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t} + \frac{mg}{k}$

$\zeta = 1$ $x(t) = (C_1 + C_2 t) e^{-\omega_n t} + \frac{mg}{k}$

$\zeta < 1$ $x(t) = e^{-\zeta\omega_n t} (A \cos \omega_d t + B \sin \omega_d t) + \frac{mg}{k}$

$$= X e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) + \frac{mg}{k}$$

Initial conditions: $x(0) = 0$ and $\dot{x}(0) = v_0 = \sqrt{2gh}$

11.20 Cramer's rule

For the equation of motion:

$$\begin{bmatrix} 2k - m\omega^2 & -k \\ -k & 2k - m\omega^2 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ 0 \end{Bmatrix}$$

For X_1 , replace the **first column** of the matrix in the numerator with the force vector:

$$X_1 = \frac{\begin{vmatrix} F_1 & -k \\ 0 & 2k - m\omega^2 \end{vmatrix}}{\begin{vmatrix} 2k - m\omega^2 & -k \\ -k & 2k - m\omega^2 \end{vmatrix}} = \frac{F_1(2k - \omega^2 m)}{(2k - \omega^2 m)^2 - k^2}$$

For X_2 , replace the **second column** of the matrix in the numerator with the force vector:

$$X_2 = \frac{\begin{vmatrix} 2k - m\omega^2 & F_1 \\ -k & 0 \end{vmatrix}}{\begin{vmatrix} 2k - m\omega^2 & -k \\ -k & 2k - m\omega^2 \end{vmatrix}} = \frac{F_1 k}{(2k - \omega^2 m)^2 - k^2}$$

11.21 General steps

1. Find the equation of motion using Newton's 2nd law for all the masses involved with the x or θ terms on the left-hand side:

$$\text{Forces: } m\ddot{x} = \sum F$$

$$\text{Moments: } J_o\ddot{\theta} = \sum M_o$$

2. Assume the steady state solution is one of the following:

$$\text{Forces: } x(t) = X \sin(\omega t - \phi), \quad \ddot{x} = -\omega^2 x$$

$$\text{Moments: } \theta(t) = \Theta \sin(\omega t - \phi), \quad \ddot{\theta} = -\omega^2 \theta$$

11.21.1 Natural frequency of 2-DOF systems

1. Always assume that the displacement of the current is greater than the displacement of the other mass, i.e. if the current mass' displacement is x_1 , assume $x_1 > x_2$.
2. Set the determinant of the matrix of the equation of motions to 0 to find the characteristic equation.
3. Solve for the roots of the characteristic equation to get the natural frequencies.

11.21.2 Mode shapes of 2-DOF systems

1. Use any of the equations of motion and make the amplitude ratio $\left(\frac{X_1}{X_2} \text{ or } \frac{X_2}{X_1}\right)$ the subject.
2. Substitute the natural frequency into the amplitude ratio equation.
3. Set either X_1 or X_2 as 1, then calculate the other value to form the mode shape vector.

12 Maths

12.1 Solutions to $\sin \theta = c$

$$\alpha = \arcsin c$$

$$\theta = \begin{cases} 2k\pi + \alpha \\ (2k+1)\pi - \alpha \end{cases} \text{ where } k = 0, 1, 2, 3, \dots$$

$$\theta = \alpha, \pi - \alpha, 2\pi + \alpha, 3\pi - \alpha, \dots$$

12.2 Small angle approximation

$$\sin \theta \approx \theta$$

$$\cos \theta \approx 1$$

$$\tan \theta \approx \theta$$

12.3 Derivatives

Chain rule:

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

Product rule:

$$\frac{d}{dx}(u \cdot v) = \frac{du}{dx} \cdot v + u \cdot \frac{dv}{dx}$$

$$\text{Quotient rule: } \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

Standard derivatives:

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arccos x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

12.4 Integrals

$$\int \frac{1}{x} dx = \ln|x|$$

$$\int \sin x dx = -\cos x$$

$$\int \cos x dx = \sin x$$

$$\int \tan x dx = \ln|\sec x|$$

$$\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\left(\frac{x}{a}\right)$$

$$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$$

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$$

$$\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln|\sqrt{x^2-a^2} + x|$$

$$\int \sec x dx = -\ln|\sec x + \tan x|$$

$$\int \csc x dx = -\ln|\csc x + \cot x|$$

$$\int \cot x dx = \ln|\sin x|$$

Integration by parts:

$$\int u dv = uv - \int v du$$

Preference for u :

1. L: log or ln.
2. I: Inverse trigonometric functions.
3. A: Algebraic functions.
4. T: Trigonometric functions.
5. E: Exponential functions.

12.5 Trigonometric identities

Quotient identities:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Reciprocal identities:

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Pythagorean identities:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\csc^2 \theta - \cot^2 \theta = 1$$

Even/odd identities:

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$